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RESEARCH ARTICLE

EG-VAN: A Global and Local Attention-Based Dual-Branch Ensemble Network With Advanced Color Balancing for Multi-Class Skin Cancer Recognition

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ABSTRACT The accurate diagnosis of skin cancer remains challenging due to factors like intra-class variation and inter-class similarity in dermoscopic images. Despite their sophistication, existing methods struggle to generalize well across diverse cases, and their computational complexity often limits real-world applicability. To address these challenges, we introduce a novel dual-path architecture in the Efficient Global-Vision Attention-Net (EG-VAN) for early detection of skin cancer. The dual-path design uniquely combines (1) EfficientNetV2S, which processes multi-scale features with high efficiency, and (2) a modified ResNet50 enhanced with a Spatial-Context Group Attention (SCGA) module and Non-Local Block. The SCGA module captures local-contextual relationships in horizontal and vertical directions to focus on subtle lesion details, directly addressing intra-class variation, while the Non-Local Block captures long-range dependencies to resolve inter-class similarity. This synergistic dual-path fusion ensures comprehensive feature extraction, enabling EG-VAN to outperform single-path architectures. Further, we introduce a Multi-Scale Feature Fusion (MFF) module to aggregate hidden features across scales, improving generalization across skin cancer types. Additionally, our pioneering color balancing method, inspired by Gray World and Retinex theory, recalibrates color distribution to highlight lesion regions, enhancing the dual-path network's input quality. We validate our EG-VAN through extensive experiments on the 9-class skin cancer dataset. Our experiments show improvement in performance by achieving an accuracy of 98.20%, a recall of 95.74%, and an F1-Score of 96.68%.

INDEX TERMS Deep learning, medical image diagnosis, color balancing, skin cancer, computer-aided diagnosis.

I. INTRODUCTION

Human skin is the most significant part of human body which provides protection and beauty as well. When Human body cells affected due to external and sometimes internal factors which results in skin cancer. Tumors emerge when skin cells

begin to grow atypically [1], [2]. Different types of skin cancer exist based on the cells that develop cancerous in any part of the skin but melanoma is the most prevalent type of skin cancer. According to the World Health Organization [3], the global number of reported skin cancer cases reached approximately 1.5 million in 2020. Among these cases, 325,000 were classified as melanoma, a highly dangerous form of skin cancer, resulting in a concerning 57,000 deaths. If this

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trend persists, it is projected that by 2040, the incidence of new melanoma cases and the mortality rate may increase by up to 50% [4]. Skin condition diagnosis relies heavily on specialist medical imagery known as dermoscopic pictures [5], [6]. Dermatologists can inspect the skin more thoroughly by utilizing a dermatoscope, a portable equipment, to take these pictures. Dermoscopy uses a magnifying lens and a light source to examine the skin's surface and deeper layers. It is sometimes referred to as dermatoscopy or epiluminescence microscopy. Dermoscopic photos help to provide a more thorough analysis of the form, color, and texture of skin lesions. Furthermore, pigmentation patterns, blood vessel existence, and other minute signs critical to a precise diagnosis can be seen in these images.

The quest to diagnose skin cancer accurately is a vital field of study. The key to successful treatment and enhancing the chances of survival lies in detecting the condition early on. While traditional diagnostic methods have been the mainstay for years, the advent of advanced techniques such as deep learning [7], [8] and transfer learning [9], [10], [11] has opened up new possibilities. These innovative approaches are making strides in refining the accuracy and expediting the process of diagnosing skin cancer, marking a significant leap forward in medical science [12], [13], [14]. Deep learning is a sort of machine learning that uses neural networks [15], [16]. These networks can be trained on huge datasets of images of skin lesions to find patterns unique to various types of skin cancer. Convolutional neural networks (CNNs), a specific kind of neural network, have demonstrated exceptional effectiveness in image recognition tasks and are hence suitable for the diagnosis of skin cancer. CNNs can increase the precision of diagnosis and expedite the procedure by recognizing important aspects including texture, color, and form [17]. Further, integrating the attention mechanism with convolutional neural networks (CNNs) can increase the model's ability to capture meaningful information in medical images [18], [19], [20].

Despite the advancement in deep learning models and attention mechanisms, a skin cancer diagnosis is still a challenging task in terms of intra-class variation and inter-class similarity. In intra-class variations, where lesions of the same type like melanoma, can vary significantly in size, color, shape, and texture due to factors like growth stage, patient demographics, or environmental conditions. This variability makes it difficult for models to generalize across different instances within the same class. On the other hand, inter-class similarity poses another challenge, as benign lesions can often appear visually similar to malignant ones making it difficult for both clinicians and deep learning models to differentiate between them. This similarity increases the risk of misclassification, especially in borderline cases. Class imbalance is another common issue in medical images like the skin cancer dataset, most of the classes greatly outnumber other classes in the same dataset. Furthermore, many existing datasets are small and may lack diversity in skin types and

lesion appearances further hindering model generalization. The presence of noise in dermoscopy images, such as hair, lighting variations, and imaging artifacts, adds another layer of complexity, as these imperfections can obscure important lesion features and reduce diagnostic accuracy. Moreover, while advanced models incorporating attention mechanisms or transformers improve accuracy, they often come with increased computational complexity, making them less feasible for real-time clinical use, especially in resource-limited settings. Finally, deep learning models are prone to overfitting when trained on small or imbalanced datasets that leading to poor generalization to unseen cases.

To address these challenges, we propose the Efficient Global-Vision Attention-Net (EG-VAN), a multi-faceted network that includes a Dual-Branch Ensemble (DBE) Network with a Multi-Scale Feature Fusion (MFF) Module. In the DBE network, the modified ResNet50 branch effectively captures both local and global features in skin images by incorporating the Non-Local Attention Block alongside the Spatial-Context Group Attention (SCGA) module. Simultaneously, the EfficientNetV2S branch leverages pre-trained weights to extract essential information. Additionally, the MFF module strengthens the model's ability to analyze various skin lesion types by integrating fine-grained and coarse-grained features. To further handle class imbalance, we employ data augmentation techniques and the focal cross-entropy loss function. Focal cross-entropy loss instead of categorical cross-entropy loss ensures the model focuses on harder-to-classify examples, thereby improving performance on underrepresented classes. Importantly, despite the addition of modules such as SCGA, the Non-Local Attention Block, and the MFF Module, the EG-VAN model remains computationally efficient without increasing the parameter count. For image denoising, we introduce a novel color-balancing technique that enhances image quality and emphasizes the disease-affected areas. We evaluate the proposed method on publicly available skin lesion datasets, including HAM10000 and a two-class skin cancer dataset containing over 10,000 dermoscopy images and nine types of skin diseases. Comparative analysis with state-of-the-art methods and extensive ablation studies demonstrates the effectiveness of our framework.

The main contributions are summarized as follows:

- We present novel EG-VAN architecture that is specifically designed for skin cancer diagnosis, combining a Dual-Branch Ensemble Network with cutting-edge attention mechanisms to capture detailed local and global features.
- We are the first to enhance ResNet50 with a Spatial-Context Group Attention (SCGA) module and Non-Local Attention Block to address intra-class variation and inter-class similarity, while maintaining computational efficiency.
- Our MFF module effectively integrates multi-scale information, ensuring superior feature representation

and improving model generalization across diverse skin lesions.

- We introduce a new color balancing method based on the Gray World algorithm and Retinex theory, which enhances dermoscopic images and highlights critical disease areas for better diagnostic accuracy.
- We tackle the challenge of class imbalance with focal loss, ensuring improved sensitivity to minority classes and minimizing misclassification risks.
- The nine-class skin cancer classification method is introduced using an aggregated dataset derived from the currently utilized data. This approach enhances the accuracy of the skin cancer categorization system, achieving an impressive accuracy score of 98.20%.

After the opening remarks, this paper takes a closer look at the body of research on detecting skin cancer through deep learning, discussing key developments and significant studies. The next section changes direction to outline the tools and techniques that form the foundation of this study, providing a clear explanation of how the EG-VAN model operates, one piece at a time. Then, just before the conclusion, there are findings of this research study in detail, examining the effectiveness of EG-VAN through different performance measures and highlighting its capabilities in diagnosing skin cancer.

II. RELATED WORK

The detection and classification of skin lesions for diagnosing skin malignancy pose challenges due to complex features and limited training datasets [21], [22], [23], [24], [25]. Obayya et al. [26] developed a new architecture MAFCNN-SCD for the early diagnosis of multi-class skin cancer problems. They applied the DBN method for the detection and classification of skin cancer from Dermoscopic images. The proposed model MAFCNN-SCD achieved 92.22% and 99.34% accuracy on the HAM 10000 and ISIC 2017 datasets respectively. Preprocessing steps, such as enhancing the diseased areas in skin cancer images, are crucial. Khurram et al. [27] introduced the Gray World algorithm to improve overall image quality, thereby enhancing the model's classification capability. Alwakid et al. [28] introduced the ESRGAN method to improve the HAM10000 dataset image quality and then the augmentation method to balance the multi-class data. The proposed Modified ResNet50 architecture obtained 86% accuracy on the HAM10000 dataset. This Study [29] highlighted the effectiveness of combining spatial and frequency domain feature extraction by applying the dual-autoencoder network. This model leverages attention modules and extreme learning machines (ELM) for robust classification, demonstrating improved accuracy and performance across different skin cancer datasets. Karthik et al. [30] propose a hybrid deep learning model for skin cancer classification, integrating Swin Transformer and Dense Group Shuffle Non-Local Attention (DGSNLA) networks. Their approach enhances

feature extraction by combining transformer-based global features with CNN-based local features, achieving high accuracy on the HAM10000 dataset. Further, Ramamurthy et al. [31] presented a deep learning model that integrates localized, contextual, and hierarchical features for skin lesion classification. The model utilizes a modified DenseNet-169 with a Coordinate Attention Module (CoAM) and a custom CNN with a Feature Pyramid Network (FPN) and Global Context Network (GCN), achieving improved classification performance.

Wang et al. [32] introduced the concept of knowledge distillation (KD) as a solution for adapting lightweight models (ResNet50, MobilenetV2), proposing a novel method called SSD-KD for skin disease classification. This method mixed intra-instance relational feature representation with existing KD research and applied on the ISIC-2019 dataset. This method achieved an 85% accuracy score for 8 skin diseases. On the other hand, Natha et al. [33] applied max voting scheme by using the traditional machine learning algorithms to improve the classification accuracy of the 7-class skin cancer dataset. Qian et al. [34] modified the original ResNet50 model and employed a multi-scale attention block in each block of the ResNet50 model. The model achieved 91.6%, 97.1% accuracy, and AUC score on the multi-class HAM10000 dataset. Wu Di et al. [35] introduced hybrid approach-based ECRNET that is spatially designed to extract both local and global information for skin cancer datasets. ECRNet utilizes coordinate attention and channel attention through the CCPA block to enhance the network's ability to capture critical image features. This method achieved 92.12% accuracy on the 7-class dataset. Vidhyalakshmi et al. [36] proposed a hybrid model combining CNN with bidirectional LSTM and Flash Butterfly Optimization for skin disease classification. Their model addresses class imbalance issues through augmentation and achieves a classification accuracy of 96.3% on the HAM10000 dataset. Ramamurthy et al. [37] introduce a two-stage network for skin disease detection using atrous residual convolutional networks. Their model incorporates segmentation and classification stages, improving inter-class performance and achieving an accuracy of 89.27% on HAM10000.

Tan et al. [38] introduced the DGLA-ResNet50 model, which integrates vertical and horizontal attention mechanisms for capturing global features and utilizes multiple convolutional layers for capturing local features. Additionally, they employed a Dual Branch input scheme to address image variety in multi-class skin cancer datasets. This approach achieved an accuracy of 90.71% with 104.2 million parameters. Lan et al. [39] proposed an improved FixCaps model with an attention mechanism that used large kernel size in the model opposite to traditional deep learning models. The proposed model obtained 96.49% accuracy. In this paper [40], a new method for skin lesion classification using deep convolutional neural networks was proposed. The approach involved constructing an ensemble of networks that work together to assess lesions more

accurately. By considering multiple shifted versions of the test image and combining the resulting class score vectors, the proposed strategy leverages the strengths of the ensemble networks and mitigates individual classification failures. Experimental results achieved an overall 83.6% accuracy on the HAM10000 dataset. Omeroglu et al. [41] introduced a multi-model scheme with soft attention mechanism based modified version of Xception model for precise multi-label skin cancer problem. The proposed framework achieved an average accuracy of 83.04% on SPC dataset. Rezaee et al. [42] introduced the self-attention based deep learning model. They used ensemble strategy with resnet18 and transformer model to extract both local and global features to further refine the model to detect skin lesion types. The proposed model achieved 97% accuracy on the 8-class dataset.

Qian et al. [43] introduced a novel approach that included three key sub-modules in the ResNet101 model. The first self-interacting attention pyramid (SAP) module consisted of channel and spatial attention. The Second Global Average Algorithm (GAA) consisted of two fully connected layers with a global average pooling layer. The third cross-layer bilinear-trilinear pooling operation consisted of average pooling layer with convolutional layers. The proposed SPCB-Net achieved 97.10% accuracy on the HAM-10000 dataset. Rehman et al. [44] introduced a weighted ensemble strategy that combined the average weighted score of five pre-trained deep learning models to predict the multi-class skin cancer problem and the proposed ensemble model achieved 93% accuracy. Recently, leveraging deep learning for feature extraction combined with traditional machine learning for classification has become a prevalent method for detecting skin cancer in medical imaging [45].

A. RESEARCH GAPS

While prior studies have advanced skin cancer diagnosis using deep learning, critical gaps persist:

Overreliance on Conventional Architectures: Existing works predominantly employ architectures like MobileNet, ResNet, and Inception, which, despite their widespread adoption, exhibit limited accuracy compared to modern, efficiency-optimized models (e.g., EfficientNetV2). These frameworks also struggle to balance computational efficiency with multi-scale feature extraction.

Narrow Strategy Adoption: Most approaches adopt a singular strategy either ensemble learning or attention mechanisms but fail to synergize both. Ensembles often combine architecturally similar models leading to redundant feature extraction and suboptimal generalization.

Neglect of Input Quality Enhancement: Few works address the inherent challenges of dermoscopic images, such as uneven illumination, artifacts, or low lesion contrast. Ignoring these issues undermines feature extraction, particularly in disease-affected regions.

B. RESEARCH CONTRIBUTIONS

Research contributions towards the proposed work are as follows.

Dual-Path Architecture: We integrate EfficientNetV2S (for multi-scale efficiency) with a modified ResNet50 enhanced by a Spatial-Context Group Attention (SCGA) module and Non-Local Block, enabling simultaneous local-contextual and global dependency modeling. This hybrid design outperforms single-strategy approaches.

Heterogeneous Ensemble: Unlike prior works, our dual-path model combines architecturally distinct networks (CNN + attention-augmented CNN), ensuring diverse and complementary feature extraction.

Color Balancing Preprocessing: We introduce a novel color correction method, inspired by Gray World and Retinex theory, to recalibrate lesion regions and enhance discriminative features.

III. MATERIALS AND METHODS

This section describes the multiple steps involved in addressing a multi-class skin cancer problem. The steps are as follows:

A. DATA PREPROCESSING

Various data preprocessing techniques were employed on the HAM10000 dataset before conducting additional experiments. Initially, a hair removal algorithm was implemented on the multi-class skin datasets to eliminate artifacts present in the images, as illustrated in “Figure 1”. This is important because the presence of hair obstructions on the lesions can influence the diagnosis and analysis of skin issues [46]. Let $f(x, y) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is the original color image with RGB 3 channels at pixel (x, y) , $g(x, y) : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the grayscale image at pixel (x, y) and $i : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the desired output color image. In starting, we defined the convolution operation $(g * s)(x, y)$ that applies a filter s to an image g at a pixel (x, y) . It is written as:

$$(g * s)(x, y) = \sum_{u=-m}^m \sum_{v=-n}^n g(x-u, y-v) s(u, v) \quad (1)$$

where s is a matrix of size $(2m+1) \times (2n+1)$ that contains the coefficients or weights of the filter. In the next step, another mathematical procedure is called morphological opening $(g \circ s)(x, y)$ that applies the erosion followed by the dilation to an image g using a structural element s . It is written as:

$$(g \circ s)(x, y) = ((g \ominus s) \oplus s)(x, y) \quad (2)$$

where $(g \ominus s)(x, y)$ is the erosion operation that preserves the shape of the structural element and removes pixels that do not fit inside it. Similarly, $(g \oplus s)(x, y)$ is the dilation operation that preserves the shape of the structural element and adds pixels that fit inside it. Both are written as:

$$(g \ominus s)(x, y) = \min_{u,v \in s} g(x+u, y+v) \quad (3)$$



FIGURE 1. (A) Original Images (B) After applying hair removal algorithm.

$$(g \oplus s)(x, y) = \max_{u, v \in s} g(x - u, y - v) \quad (4)$$

where s is a set of pixels that defines the shape and size of the structural element. The third step Black top-hat filtering $(g - g \circ s)(x, y)$ enhances small bright structures on a dark background. It involves taking the difference between the input image and the result of the opening operation with a structural element. It is written as:

$$(g - g \circ s)(x, y) = g(x, y) - (g \circ s)(x, y) \quad (5)$$

where $\circ$ represents the opening operation. Another step involves thresholding the result of the previous step to obtain a binary image where the hair/marking contours are intensified. The threshold operation is represented mathematically as follows:

$$t(x, y) = \begin{cases} 1 & \text{if } (g - g \circ s)(x, y) < T \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

The final step Inpainting is the process of filling in missing or damaged regions in an image using information from the surrounding areas. For this purpose, the Telea inpainting algorithm is used that minimizes the difference between the original image and the inpainted image while preserving edges and textures.

$$i(x, y) = \operatorname{argmin}_{i \in I} \left(\int_{\Omega/M_t} |i - f|^2 dx + \gamma \int_{\Omega} |\Delta i|^2 dx \right) \quad (7)$$

where $i(x, y)$ is the inpainted image at pixel location (x, y) , Ω is the data fidelity term that measures the difference between the inpainted image and the original image, M_t is the mask that specifies the region to be inpainted, and γ is a regularization parameter that controls the smoothness of the inpainted image.

After performing the hair removal process on the HAM10000 dataset, we adopted a novel preprocessing approach to mitigate the challenges arising from poor illumination conditions and color variation in Dermoscopic images. Our methodology incorporated a fusion of the Gray World algorithm with Retinex theory, resulting in marked enhancements in contrast and lighting within key areas of the images. The Gray World algorithm was utilized to correct color deviations by adjusting the color balance. Concurrently, Retinex

theory was applied, based on the principle that the visibility of an object is influenced by its reflectance in relation to the ambient light, to improve image clarity. The mathematical integration of the Gray World algorithm and Retinex theory is articulated as follows:

$$\begin{bmatrix} R_{avg} \\ G_{avg} \\ B_{avg} \end{bmatrix} = \frac{1}{N} \sum_{i=1}^N \begin{bmatrix} R_i \\ G_i \\ B_i \end{bmatrix} \quad (8)$$

$$\begin{bmatrix} \alpha_R \\ \alpha_G \\ \alpha_B \end{bmatrix} = \begin{bmatrix} G_{avg} + B_{avg}/2R_{avg} \\ 1 \\ R_{avg} + G_{avg}/2B_{avg} \end{bmatrix} \quad (9)$$

$$I_{filtered}^S = G_s * I_{RGB} \quad (10)$$

$$L^S = \log(I_{filtered}^S) - \log(I_{rgb}) \quad (11)$$

$$I_{retinex}^{RGB} = \sum_{s=1}^S W_s \times L^s \quad (12)$$

$$I_{combined} = I_{retinex}^R \times \alpha_R + I_{retinex}^G \times \alpha_G + I_{retinex}^B \times \alpha_B \quad (13)$$

where $I_{combined}$ denotes an enhanced RGB image, I_{RGB} represents an original RGB Image, N signifies the total number of pixels in the image, S stands for the total number of scales, considered, α_R , α_G , and α_B denote scaling factors ensuring the correction and balancing of color channels based on the average color values of the image. G_s denotes a Gaussian kernel of scale S , L^S represents the log-difference image at scale S , and W_s represents the weight given to the log-difference image at scale s . In “equation 8 and 9” we first applied Gray World algorithm on 9-class dataset to measure the average illumination level of the dermoscopic image. This technique is predicated on the idea that the average hue in a well-lit scene is typically neutral or gray. The program can correct for color imbalances in the image and produce a more natural and balanced color representation by using this assumption.

In order to improve the skin lesion quality even more, we then applied the Retinex theory concepts to “equation 10 to 13”. By dissecting an image into its illumination and reflectance components, Retinex theory enables us to separate the effects of lighting from the inherent color characteristics of the skin lesion. Fine details are emphasized and the effect of changing illumination is lessened by this breakdown, resulting in visually appealing and consistent photographs. We first apply spatial filtering to the image, and then we compute the log-ratio to achieve dynamic range compression and contrast augmentation. The minor details in the skin lesions are more visible because to this procedure, which highlights the variations in relative luminance between pixels. Additionally, multiple log-ratio images are calculated using different spatial filters, and each log-ratio image is weighted based on its influence. The weighted log-ratio images are then combined to obtain the Retinex component of the image.

Finally, the Retinex-corrected image is generated by integrating the individual Retinex components for each color channel using the previously computed scaling factors. This approach produces Dermoscopic skin cancer images with improved color consistency, enhanced contrast, and reduced

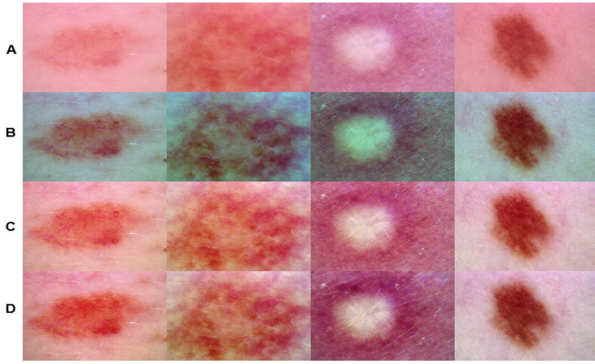


FIGURE 2. (A) Original Image (B) gray world algorithm generated images (C) Retinex theory generated images (D) combined gray world algorithm + Retinex theory generated image.

sensitivity to variations in illumination conditions, as shown in “Figure 2”.

We also included the crop function to eliminate the need-less part of the image. Finally, we set the image size according to our model. The crop function is defined mathematically as:

$$\text{crop}(x, y) = \begin{cases} 1, & \text{if } |x - c_x| \leq \frac{s}{2} \text{ and } |y - c_y| \leq \frac{s}{2} \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

where c_x and c_y is the coordinates of the center point and is the crop size. The “Equation 14” takes two inputs and uses the absolute value function, denoted by $| |$, to measure the distance between the inputs and the center point along each axis. If both distances are less than or equal to half of the crop size, then the inputs are inside the region and the function returns 1. Otherwise, the inputs are outside the region and the function returns 0.

B. DATA AUGMENTATION

Data augmentation is a technique that creates new data from existing data by applying random transformations. This can help machine learning and computer vision models to learn from more diverse and robust data and reduce over-fitting. It was also necessary because in the HAM-10000 dataset some of the classes had more images than other classes. So, during training a large of images from certain classes learn more features compared to classes with fewer images, which is not ideal for a good model. To address this issue, Multiple Data augmentation techniques were employed on the HAM-10000 dataset. Let's assuming $\sum_{i=1}^N (x_i, y_i)$ are a dataset of input-output pairs, where x_i is the input data and y_i is the associated label, and we want to perform data augmentation on x_i , augmentation function $f(x_i)$ described as follows:

$$f(x_i) = \sum_{j=1}^{20} T_j(x_i) \quad (15)$$

Here, $f(x_i)$ represents the function applied to the input x_i . The sum operator $\sum$ signifies the summation over 20 different transformations denoted by $T_j(x_i)$, where j ranges from 1 to 20. Each T_j represents a specific transformation

applied to x_i , and the sum indicates that these transformations are combined by addition. After applying these augmentation techniques, we produced new input data $f(x_i)$ that are identical to the original data but with some changes by putting these random modifications in a particular order.

C. OVERVIEW OF PROPOSED MODEL

In this section, we provide a detailed description of our proposed Efficient Global-Vision Attention-Net (EG-VAN), which is specifically designed for dermoscopic skin cancer images. The EG-VAN model comprises four main components: the first component is a Dual-Branch Ensemble Model, which combines complementary features from different branches to improve classification accuracy. The second component is the Spatial-Context Group Attention (SCGA) Module, which is designed to capture both local and contextual information in the horizontal and vertical directions within the skin images. The third component, the Non-Local Block Attention Module, is included to model long-range dependencies and enhance feature extraction over larger spatial regions. Finally, the fourth component is the Multi-Scale Feature Fusion (MFF) Module, which effectively extracts features across multiple scales to address the similarities between different skin lesion types. Each of these components is introduced and discussed individually in the following sections. The complete proposed model is illustrated in Figure 3”.

1) DUAL-BRANCH ENSEMBLE MODEL

Ensemble modeling can provide superior performance when dealing with dermoscopic skin images, which contain high-level information that cannot be easily managed by a single model. Additionally, leveraging both local and global information can significantly enhance model performance. Convolutional Neural Networks (CNNs) primarily capture local information due to the localizability of the convolutional layer, but they struggle to align global information and long-range dependencies in skin images. The Transformer model addresses this limitation by effectively understanding global information and long-range dependencies. However, transformer models are computationally heavy and require additional resources during training. To address these challenges, we introduce a Dual-Branch Ensemble (DBE) Network, which consists of a pre-trained EfficientNetV2S [47] and a modified version of ResNet50 [48]. The EfficientNetV2S [47] was selected due to its higher accuracy compared to traditional models and its predecessor, particularly in classification tasks. Another advantage of EfficientNetV2S is its use of Fused-MB convolution layers in the earlier layers, replacing the standard MB-Conv layers, resulting in faster training. Additionally, EfficientNetV2S utilizes a fixed 3×3 window size across all layers, making it more effective at extracting meaningful features from skin images compared to EfficientNetV1 [49].

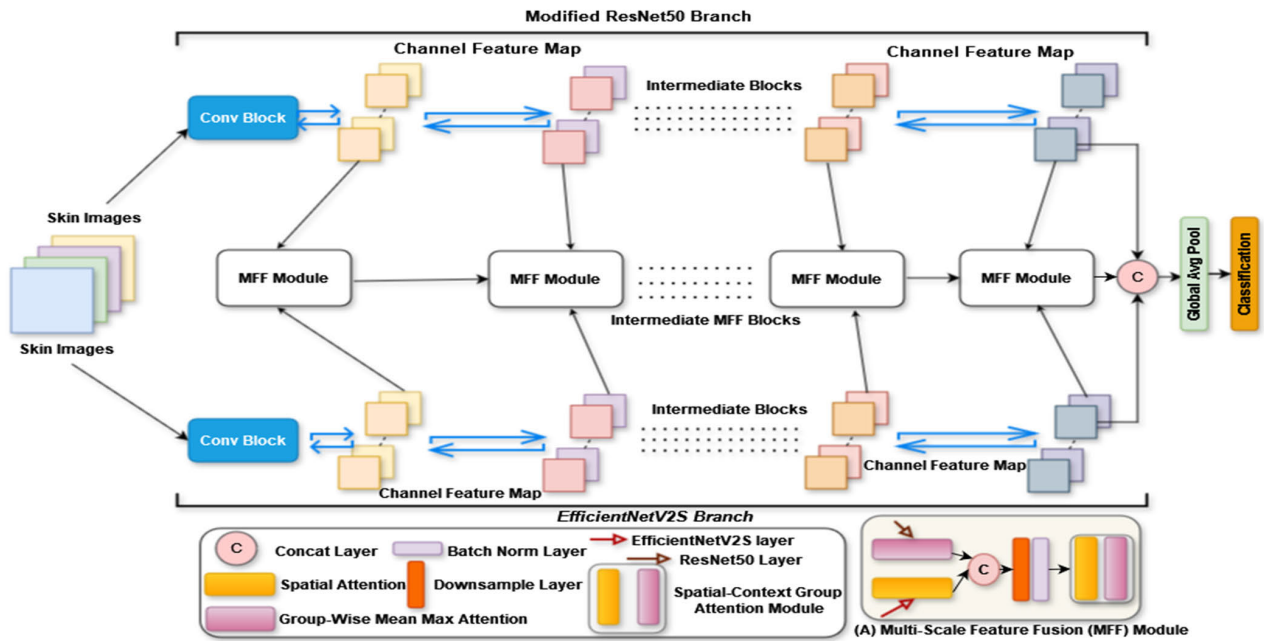


FIGURE 3. The complete workflow of proposed EG-VAN model.

For the second branch of the DBE network, we selected a modified version of ResNet50 [48]. The ResNet50 branch is enhanced with both global attention (Non-Local Block) and local attention (SCGA module) mechanisms. The SCGA module captures fine-grained spatial details by computing attention weights in horizontal and vertical directions, addressing intra-class variation. Simultaneously, the Non-Local Block captures long-range dependencies, enabling the model to understand the global context of lesions and handle inter-class similarity. These mechanisms interact within the ResNet50 branch, where the SCGA module extracts local features, and the Non-Local Block refines them by incorporating global context. This synergy allows the model to leverage both detailed and contextual information, improving its ability to distinguish between complex skin lesions. The outputs of the ResNet50 branch are then fused with the EfficientNetV2S branch through the Multi-Scale Feature Fusion (MFF) module, ensuring a comprehensive feature representation for accurate diagnosis. Despite these enhancements, the modified ResNet50 retains the same parameter count as the original model, making it computationally efficient and eliminating the need for additional resources during training. The layer structure of both models is presented in the “table 1”.

2) SPATIAL-CONTEXT GROUP ATTENTION(SCGA)

In skin image datasets, lesion size variations across multiple directions pose significant challenges, as local attention mechanisms alone prove insufficient for accurate diagnosis of suspicious areas. Additionally, intra-class variations in skin cancer images limit model capabilities for precise

classification. To address these limitations, we introduce the Spatial-Context Group Attention (SCGA) module, which enhances model performance without increasing parameters. The SCGA module employs two branches: the first branch incorporates spatial attention to capture local features in skin lesions, while the second branch utilizes Group-wise Mean-Max Attention (GMA) to detect suspicious areas in both horizontal and vertical directions. This design effectively handles multi-scale feature extraction and improves diagnostic accuracy through comprehensive contextual information processing as depicted in fig 4.

Firstly, the spatial attention applies multiple 1×1 convolutional layers with ReLU activation functions on a 2D input tensor to extract feature maps. The process begins with a 64-filter convolutional layer, followed by a 16-filter layer with a dilation rate of 2 for multi-scale context, and then an 8-filter layer. A final 1×1 convolutional layer with a sigmoid activation function is used to scale the values to a range between 0 and 1, resulting in a local attention mask that highlights significant areas of the input. To incorporate global information, the local attention mask undergoes global average pooling. The pooled result is then expanded to match the spatial dimensions of the input for broadcasting. This global information is multiplied element-wise with the local attention mask, refining the attention mechanism to consider local features. A convolutional layer with linear activation and fixed weights (initialized to ones) is then used to up-sample the refined attention mask to match the input tensor's channel dimensions. The final masked feature map X_i is created by multiplying the input tensor element-wise with this up-sampled attention mask.

TABLE 1. The first five columns present the modified ResNet50 model layer structure, while the remaining columns depict the EfficientNetV2 layer structure.

Stage	Operator	Stride	#Channels	#Layers	Stage	Operator	Stride	#Channels	#Layers
0	Conv 7 x 7	2	64	1	0	Conv 3 x 3	2	24	1
1	Max Pool 3 x 3	2	64	1	1	Fused-MBConv1, k3x3	1	24	2
2	Conv Block: 1x1, 64 → 3x3, 64 → 1x1, 256 (conv),	1	256	3	2	Fused-MBConv4, k3x3	2	48	4
	Spatial-Context Group Attention (SCGA) Module	-	256	1	3	Fused-MBConv4, k3x3	2	64	4
3	Conv Block: 1x1, 128 → 3x3, 128 → 1x1, 512 (conv)	2	512	4	4	MBConv4, k3x3, SE0.25	2	128	6
	Spatial-Context Group Attention (SCGA) Module	-	512	1	5	MBConv6, k3x3, SE0.25	1	160	9
4	Conv Block: 1x1, 256 → 3x3, 256 → 1x1, 1024 (conv)	2	1024	6	6	MBConv6, k3x3, SE0.25	2	256	15
	Non-Local Block	-	1024	1	7	Conv 1 x 1	-	1280	1
5	Conv Block: 1x1, 512 → 3x3, 512 → 1x1, 2048 (conv)	2	2048	3					
	Non-Local Block	-	2048	1					

Our GMA begins with splitting the final output of spatial attention X_i into multiple groups (G). Each group is designed to capture and process a specific set of channels within the input tensor. The idea behind this grouping is to enable the network to attend to different channel information independently. This grouping operation is achieved by dividing the channels into G equally sized subgroups.

$$X_g = X_i \left[:, :, :, g, \frac{C}{G} : (g+1) \cdot \frac{C}{G} \right] \quad (16)$$

where X_g represents the input to the g-th group, C is the total number of channels, and G is the number of groups. For each group X_g , the mechanism extracts horizontal and vertical features using two key statistics: group-wise mean μ_g and group-wise max $\max_g$. These statistics represent the aggregated information within each subgroup. The mean value captures the average information, while the max value identifies the most salient features within the subgroup. The horizontal and vertical features are computed by taking the mean and max values along the width and height dimensions:

$$\mu_g^h = \frac{1}{H} \sum_{h=1}^H X_g[h, :, :] , \mu_g^v = \frac{1}{W} \sum_{w=1}^W X_g[:, w, :] \quad (17)$$

$$\max_g^h = \max_{h=1}^H X_g[h, :, :] , \max_g^v = \max_{w=1}^W X_g[:, w, :] \quad (18)$$

These statistics provide condensed representations of the spatial features in both horizontal and vertical directions. Next,

the horizontal and vertical features of each group (F_g) are concatenated individually along the channel dimension. This creates a new tensor that combines both the average and maximum information for each group.

$$(F_g) = [\mu_1^h; \max_1^h; \dots; \mu_g^h; \max_g^h; \mu_1^v; \max_1^v; \dots; \mu_g^v; \max_g^v] \quad (19)$$

subsequently, separate convolutional operations are applied to the horizontal and vertical features using 1×1 convolutions, generating attention scores A_g^h and A_g^v for both directions. The activation function used here is the sigmoid function (σ), which ensures that the attention scores lie within the range of 0 to 1, representing the strength of attention.

$$A_g^h = \sigma((F_g^h) * f + b_h), A_g^v = \sigma((F_g^v) * f + b_v) \quad (20)$$

where $f : \mathbb{R}^{H \times W \times 2G} \rightarrow [0, 1]^{H \times W \times G}$ is a pre-defined convolutional layer with kernel size 1, $b \in \mathbb{R}^G$ is a biased term and $*$ denotes element-wise multiplication. The final attention scores (A_g) are computed by combining both horizontal and vertical attention through element-wise addition followed by a sigmoid activation. Further, Element-wise multiplication then between the original group features (X_g) and their corresponding attention scores (A_g) is performed. This operation effectively emphasizes or de-emphasizes features within each group based on their importance as determined by the attention scores. The attended group features are denoted as (X'_g).

$$(X'_g) = A_g * X_g \quad (21)$$

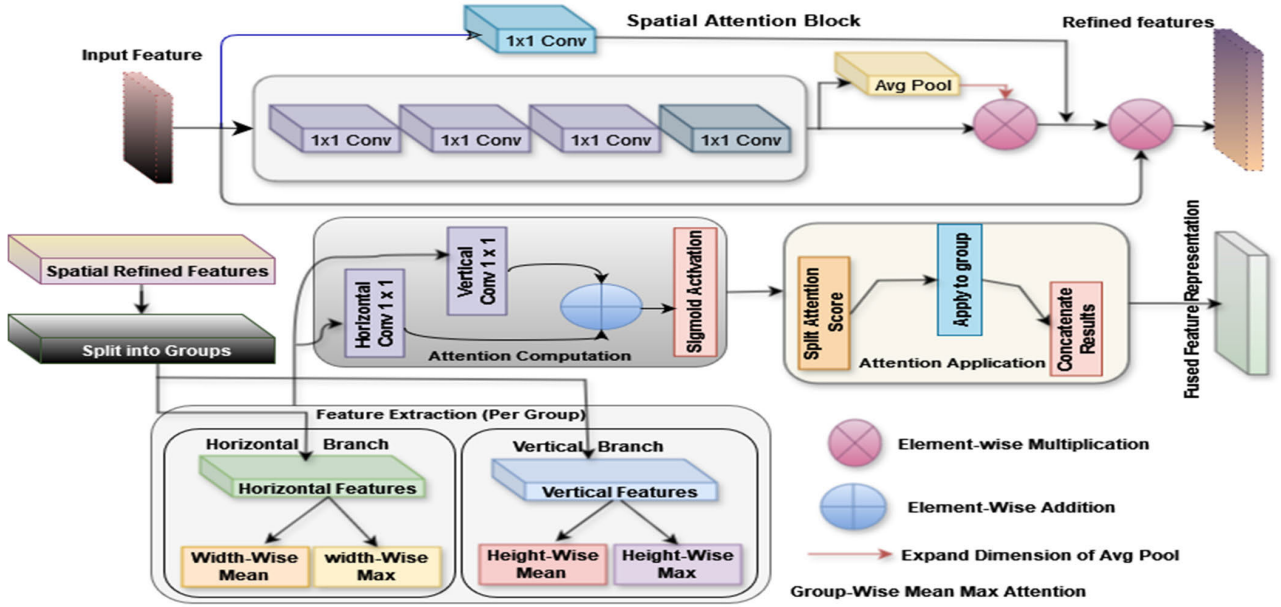


FIGURE 4. Complete overview of proposed Spatial-Context Group Attention mechanism.

Finally, the attended groups (X'_g) are concatenated along the channel dimension to form the attended input tensor (O'). This attended input tensor is a combination of all the group features, where each group's features have been individually weighted based on their relevance as determined by the attention scores.

$$O' = [X'_1, X'_2; \dots; X'_G] \in \mathbb{R}^{H \times W \times C} \quad (22)$$

3) NON-LOCAL BLOCK (NLB)

To address the limitations of traditional CNNs in modeling long-range spatial dependencies, we incorporated the Non-Local Block (NLB) [50], a self-attention mechanism tailored for image data in Figure 5. Unlike standard convolutional layers, which rely on local receptive fields, NLBs compute pairwise relationships across the entire feature map, enabling the network to capture global context and subtle inter-class variations in dermoscopic images. Unlike transformer-based self-attention, which operates on sequences and requires extensive computational resources, the Non-Local Block has been incorporated into the ResNet-50 model without substantially increasing its complexity. The core idea of the non-local block is to compute the response at a given position as a weighted sum of features from all positions in the input. For an input feature map x , the non-local operation is defined by:

$$y_i = \frac{1}{C(x)} \sum_{j'} f(x_i, x_j) g(x_j) \quad (23)$$

where i represents the index of an output position, j enumerates all possible positions in the feature map and f is a pairwise function that computes a similarity score between position i and all positions j . g is a unary function that

generates a representation of the input at the position j . and $C(x)$ is a normalization factor to ensure proper scaling.

For skin cancer classification, we used the embedded Gaussian version of the non-local block. The pairwise function f is defined as:

$$f(x_i, x_j) = e^{\theta(x_i)^T \phi(x_j)} \quad (24)$$

where $\theta(x_i) = W_\theta x_i$ and $\phi(x_j) = W_\phi x_j$ are two learned embeddings. Thus, the complete non-local operation can be written as:

$$y_i = \frac{1}{\sum_{j'} e^{\theta(x_i)^T \phi(x_j)}} \sum_{j'} e^{\theta(x_i)^T \phi(x_j)} g(x_j) \quad (25)$$

where $g(x_j) = W_g x_j$ is another learned embedding. To compute the non-local operation efficiently, we express it using matrix operations:

$$y = \text{softmax} \left(X W_\theta^T W_\phi X^T \right) g(X) \quad (26)$$

where $X \in \mathbb{R}^{N \times C}$ represents the input feature map reshaped into a matrix, with N being the number of positions and C the number of channels. The final output of the non-local block is given by:

$$z = W_z y + x \quad (27)$$

This residual connection z allows the non-local block to be seamlessly integrated into a pre-trained model without disrupting its original behavior. To improve computational efficiency, we apply spatial reduction to the key and value functions:

$$\phi(x_j) = W_\phi x_j g(x_j) = W_g x_j \quad (28)$$

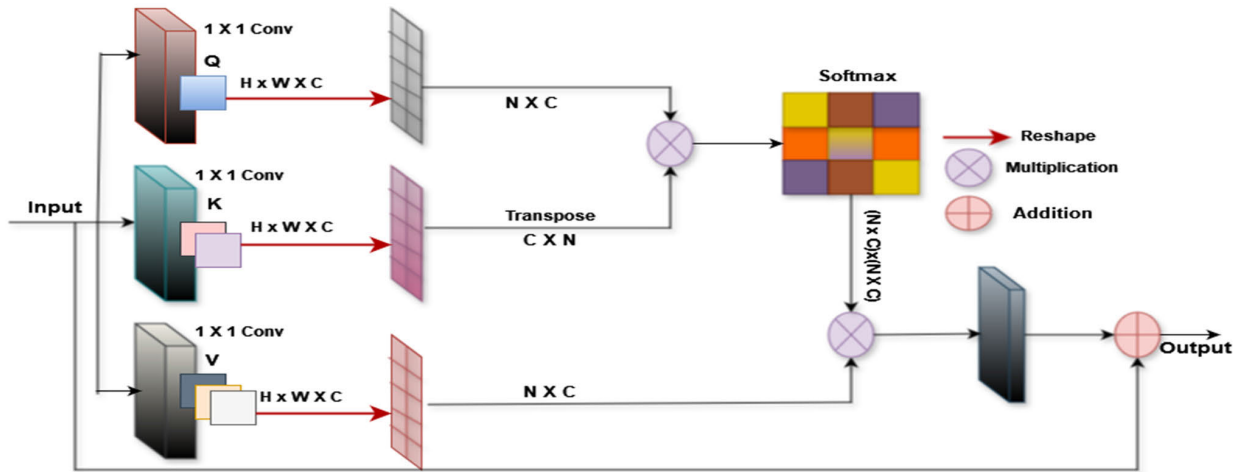


FIGURE 5. Complete overview of Non-Local Block Attention Mechanism.

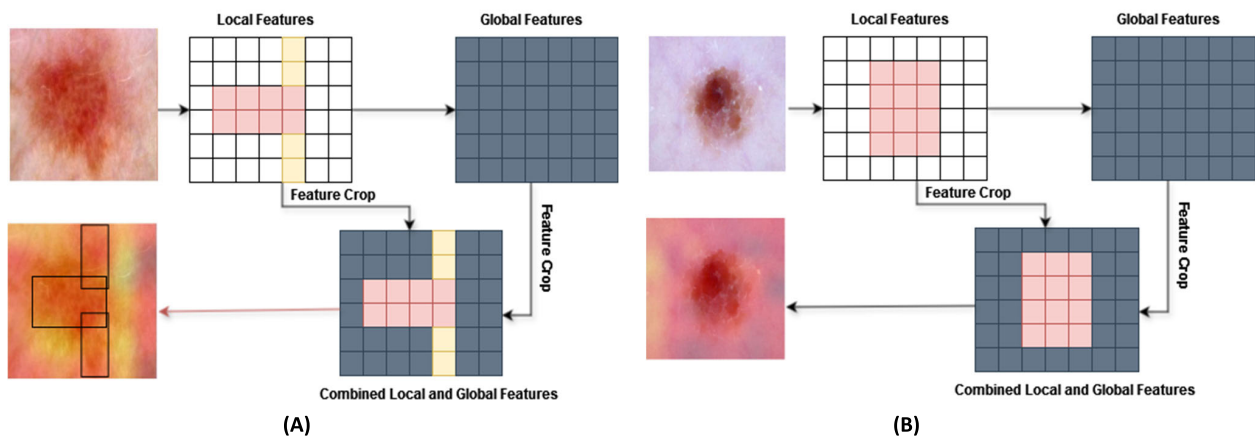


FIGURE 6. Details of two different scenarios in the modified ResNet50 model that capture both local and global information in skin cancer images. To avoid complexity, we simplified both images and did not include the complete ResNet50 model.

where j' is the index of a subsampled set of key positions. This reduction decreases the computational load while retaining the essential contextual information.

4) ADDING SCGA AND NLB IN RESNET50

The image (A) and image (B) in “Figure 6” demonstrate different approaches to feature analysis in our modified ResNet50 model, each showing how the model processes and combines local and global features for skin cancer detection. While both images follow the same general principle of feature extraction and combination, they illustrate different scenarios and analysis patterns.

In Figure 6(A), we see a cross-sectional approach to feature analysis. The initial skin lesion presents with a more diffuse, spreading pattern characteristic of certain types of skin cancers. The local feature extraction stage, handled by our SCGA modules in the first two blocks of the modified ResNet50, shows an interesting pattern where both a horizontal red section and a vertical yellow strip are highlighted. This

cross-sectional analysis allows the model to capture features along horizontal and vertical axes, potentially identifying important patterns that might not be apparent in a single direction. The red section focuses on the main lesion characteristics, while the yellow strip provides a cross-sectional view that might capture subtle variations in the lesion's structure. When these local features move through to the global feature processing stage (represented by the gray block), the non-local blocks in our model's later stages analyze how different parts of the image relate to each other. This global analysis is crucial for understanding the overall context of the lesion. The final combined feature map in the first image shows how both the red and yellow features are integrated with the global information, resulting in a comprehensive analysis that identifies multiple regions of interest, as indicated by the black boxes in the output image.

The figure 6(B), in contrast, demonstrates a more focused approach to feature analysis. Here, we see a well-defined circular lesion, and the local feature extraction focuses on

a specific concentrated area, shown by the red region in the grid. This represents how our SCGA modules can adapt to more localized, well-defined lesions. The global feature processing still considers the entire image space, but in this case, it's more about understanding how this focused region relates to the surrounding healthy skin. The combination of features in the second image shows how the model maintains focus on the central lesion while incorporating broader contextual information. The final output preserves the clarity of the original lesion while potentially enhancing important diagnostic features through the integration of both local and global information. This focused approach is particularly useful for well-defined lesions where the boundaries and internal characteristics are key diagnostic factors.

5) MULTI-SCALE FEATURE FUSION MODULE (MFF)

Classification of multi-class skin lesions is a difficult task due to the same structure of skin classes. To overcome this difficulty, we introduce a Multi-Scale Feature Fusion module illustrated in Figure 3(A) that can make the model better to understand the behavior of the skin lesion. Our MFF start starts using the Spatial Attention (SA) that is applied to the feature maps produced by the EfficientNetV2 layer. This process can be represented as:

$$SA(x) = \sum_{i=1}^H \sum_{j=1}^W f_{SA}(x_{ij}) \quad (29)$$

Here x is the input feature map, and f_{SA} is the spatial attention function that processes each element x_{ij} of the feature map. The double summation iterates over all spatial locations (i, j) of the feature map with height and width. Simultaneously, The ResNet50 model layer is processed using a group-wise mean max attention mechanism. This is described by the equation:

$$GMM(x) = \frac{1}{G} \sum_{g=1}^G \max_{k \in G_g} f_{GMM}(x_k) \quad (30)$$

In the upper equation, G represents the number of groups, and G_g is the set of elements in group g . The function f_{GMM} applies the group-wise mean max attention to each element x_k , and the maximum value within each group is taken. The mean of these maximum values across all groups is then computed.

After that, the concatenation of spatial attention and group-wise mean attention information before passing it through a separable convolutional (conv) layer with strides is innovative. This step ensures that the model captures both fine-grained and coarse-grained features. The stride value of 2 in the separable conv layer effectively reduces the spatial dimensions, performing down-sampling, which is crucial for reducing computational complexity and focusing on the most salient features. The down-sampling information is then normalized using the batch normalization layer. Finally, the normalized feature map is passed through the Spatial-Context Group Attention Module which is described in subheading 2.

6) EFFICIENT GLOBAL-VISION ATTENTION-NET(EG-VAN)

This paper introduced an Efficient Global-Vision Attention-Net (EG-VAN) that is represented as the function $f: \mathbb{R}^{H \times W \times 3} \rightarrow \mathbb{R}^K$, where K is the number of classes. The layers structure of the function is as follows:

Initially, a Dual-Branch Ensemble (DBE) network is applied. In the DBE network, both baseline EfficientNet-V2S and modified ResNet50 models were employed to analyze the input image, resulting in the generation of a feature map. Subsequently, we produced an output from the first block of both baseline models. Additionally, we incorporated a novel Multi-Scale Feature Fusion (MFF) module and passed the first output individually of both baseline models to it as mentioned in the upper subsection V. Following this, we modified the next blocks of both EfficientNetV2S and ResNet50 models and passed the outputs of each block of these models individually to the novel Multi-Scale Feature Fusion (MFF) module. In the end, we concatenated all outputs to process the final feature vector GAP. Finally, to establish a probability distribution over the classes, the vector H was passed through a dense layer incorporating a Soft-Max activation function, as illustrated in “Figure 3” of the paper.

IV. EXPERIMENTS

This section includes model prediction using different parameters. It provides an analysis of the model's performance under various settings. Additionally, the proposed model's functionality is explained with Grad-CAM analysis, which helps to understand the model's decision-making process and highlight the areas in the image that contribute most significantly to the predictions.

A. DATASET DESCRIPTION

The HAM-10000 dataset is vastly used in machine learning research that is a large-scale repository of dermatoscopic images [51], [52], [53]. Comprising over 10,000 images, it serves as a critical tool for algorithm development and benchmarking within the field of computational dermatology. This dataset is particularly notable for its diversity, encompassing a wide range of skin lesions across various diagnostic categories. It includes common benign lesions such as melanocytic nevi, as well as malignant conditions like melanoma and basal cell carcinoma. Other categories represented are actinic keratosis, benign keratosis-like lesions, dermatofibroma, and vascular lesions. Each image in the dataset is labeled with one of these diagnostic categories, providing a rich ground truth for supervised learning models.

The second skin cancer dataset [45], [54] consists of two classes: malignant and benign, containing over 3,200 images. This 2-class skin cancer dataset is also used in the existing study. By combining the HAM10000 dataset with the 2-class skin cancer dataset, we created a 9-class dataset. The complete details of the number of images in each class are provided in Table 2.

TABLE 2. Complete detail of 9-class skin cancer dataset.

Class Name	Dataset Source	Original Images	Testing Images	Train images	Augmented Images
Actinic keratoses (AKIEC)	HAM10000	327	33	294	2206
Basal Cell Carcinoma (BCC)	HAM10000	514	51	463	2537
Benign Keratosis Lesion (BKL)	HAM10000	1099	110	989	7011
Dermatofibroma (DF)	HAM10000	115	12	103	1397
Melanoma (MEL)	HAM10000	1113	111	1002	6998
Vascular Lesion (VASC)	HAM10000	142	14	128	1372
Melanocytic Nevi (NV)	HAM10000	6705	671	6034	00
Malignant	ISIC 2017	1497	150	1347	6653
Benign	ISIC 2017	1800	180	1620	6380

B. MODEL OPTIMIZATION

This paper presented a method for training an EG-VAN $f_\theta(x)$ with parameters θ to resolve a multi-class skin problem, which is a classification task involving C classes. It assumes a training dataset of input and output pairs (x_i, y_i) . The model parameters θ are initialized randomly, and the loss function $L(\theta)$ is defined using Focal Loss [55] instead of categorical cross-entropy. Focal Loss is particularly advantageous for handling class imbalance in skin cancer datasets by focusing on hard-to-classify examples. The loss function is expressed as:

$$L_{focal}(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \alpha_c \times (1 - (f_\theta(x_i) \cdot c)^\gamma) y_{i,c} \log(f_\theta(x_i) \cdot c) \quad (31)$$

Here, the indicator function $y_{i,c}$ returns 1 if the i -th training example belongs to class C and 0 otherwise. The likelihood that the i -th input belongs to class C is predicted by the expression $f_\theta(x_i) \cdot c$. The expression α_c is the balancing factor, and γ is the focusing parameter that adjusts the importance of hard examples. The loss function's gradient about the model's parameters is calculated as:

$$\nabla_\theta L_{focal}(\theta) = \frac{1}{N} \sum_{i=1}^N \alpha (1 - (f_\theta(x_i))^\gamma) (f_\theta(x_i) - y_i) \otimes x_i \quad (32)$$

where $\otimes$ denotes the outer product. The Adamax optimization algorithm is then applied. The first and second-moment vectors, m and v , respectively, are initialized as zero vectors. The hyperparameters, including the learning rate α , β_1 , β_2 , and ε , are set. The following steps are repeated until convergence or for a fixed number of iterations:

$$m = \beta_1 m + (1 - \beta_1) \nabla_\theta L_{focal}(\theta) \quad (33)$$

$$v = \max(\beta_2 v, |\nabla_\theta L_{focal}(\theta)|) \quad (34)$$

$$\hat{m} = \frac{m}{1 - \beta_1^t}, \quad (35)$$

$$\hat{v} = \frac{v}{1 - \beta_2^t} \quad (36)$$

where t represents the current iteration number, m , and v are the updated first and second-moment vectors respectively.

Similarly, $\hat{m}$ and $\hat{v}$ are computed as bias-corrected first and second-moment estimates respectively. The model parameters are updated as follows:

$$\theta = \theta - \alpha \frac{\hat{m}}{\hat{v} + \varepsilon} \quad (37)$$

The Adamax optimizer is a variant of the Adam optimizer that replaces the second-moment estimate with the infinity norm in the update step. Specifically, step 4 is replaced with:

$$u = \max(\beta_2 v, |\nabla_\theta L_{focal}(\theta)|_\infty), v \leftarrow u \quad (38)$$

C. IMPLEMENTATION DETAIL

We first divided the HAM-10000 dataset into training and testing. The testing dataset consisted of 1332 original images. To avoid the overfitting problem, we applied data augmentation technique on training dataset to improve the all the classes in the 9-class dataset. After that, we further divided the training dataset into training and validation. The training dataset contained 37,883 images that consisted of original and augmentation images. Similarly, the validation dataset contained 6,686 images that also consisted of original and augmentation images. For training purposes, the epoch value was set to 25 for all models. Similarly, the batch size value was set to 32 and initially, the learning rate value was set to 0.001 with adamax optimizer. The patience and stop patience values were set to 1 and 3 respectively on learning rate. If the validation loss does not improve after one epoch, reduce the learning rate by 0.5. We set the gamma and alpha values 2 and 0.25 for focal loss respectively. All the experiments were performed by using the 3.10 Python version with open source TensorFlow and Pytorch libraries on a GPU-optimization workstation. The validation dataset is used to save the best results. Our model has been evaluated based on various metrics including accuracy, precision, recall, F1-Score, and loss.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (39)$$

$$Precision(P) = \frac{TP}{TP + FP} \quad (40)$$

$$Recall(S) = \frac{TP}{TP + FN} \quad (41)$$

$$F1 - Score = 2 \frac{(P \times S)}{(P + S)} \quad (42)$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

D. RESULTS AND ANALYSIS

In this section, we perform a series of comprehensive experiments to evaluate the effectiveness of the proposed model and its components.

1) ANALYSIS OF THE PROPOSED NETWORK

This table 3 presents a detailed performance comparison of the proposed EG-VAN model applied to two different skin cancer datasets. In the 9-class dataset, the model shows excellent performance across most classes. For AKIEC, it achieves 98.20% accuracy but with a relatively lower recall of 75.76%, indicating some cases were missed. BCC has near-perfect metrics, with 99.77% accuracy and a balance between precision and recall at 98.04%. For BKL, the model achieves 99.25% accuracy with a high F1-score, reflecting balanced precision and recall. DF and VASC classes see perfect performance, with all metrics reaching 100%, meaning no cases were misclassified. MEL has slightly lower recall 92.79% but maintains high precision and overall accuracy at 98.20%. NV, one of the most common classes, sees very high performance with a recall of nearly 100%, indicating the model detects most cases accurately. Additionally, the benign and malignant categories show overall accuracy near 99.85%, demonstrating strong performance in distinguishing between non-cancerous and cancerous lesions.

In the 7-class dataset, the model's performance improves slightly in some categories. For AKIEC, accuracy rises to 99.10%, and recall increases, suggesting better detection of difficult cases. BCC also performs exceptionally well with a perfect recall, meaning all cases were correctly identified. BKL and NV remain consistent with their high scores across both datasets, while DF shows a slight drop in recall to 91.67%, which affects its overall F1 score. MEL also sees a minor improvement in accuracy compared to the 9-class dataset, with precision and recall remaining strong. Like in the 9-class dataset, VASC maintains perfect scores, indicating the model's reliability for this class.

“Figures 7 and 8” present the confusion matrix and ROC-AUC curve results of both skin cancer datasets for proposed EG-VAN. A confusion matrix is a tabular representation that compares the actual classifications of a dataset with the predictions made by a model. It is typically divided into four true positives false positives, true negatives, and false negatives quadrants. The true positive and true negative values indicate how accurately the model made correct predictions, on the other side false positives and false negatives highlight the model's errors. By examining the distribution of these values, we can gain insight into the model's strengths and weaknesses. This detailed breakdown helps identify areas where improvements can be made, providing a clear understanding of the model's overall effectiveness. For 9-class

skin dataset in figure 7(a) of confusion matrix, EGVAN only misclassified 24 cases in which most of the cases were belong to AKIEC and Melanoma classes. On the other hand, our EG-VAN gave excellent result against DF, VASC and malignant classes and predict all the cases within their particular classes. For 7-class skin dataset in figure 8(a), EG-VAN misclassified total of 22 cases in which most of the cases were belong to MEL, and AKICE classes. while our model demonstrated highly accurate predictions for the VASC and BCC classes, consistently classifying cases correctly within their respective categories. The ROC-AUC curve further establishes the effectiveness of our EG-VAN model. The ROC (Receiver Operating Characteristic) curve is a graphic representation that compares the true positive rate (TPR) and false positive rate (FPR) for different classifier levels. The area under the ROC curve (AUC), with a value of 0.5 denoting a fully random classifier and a value of 1.0 denoting a perfect classifier, is frequently used as a summary statistic to measure the overall performance of the classifier. For 7-class and 9-class datasets in Figures 7(b) and 8(b), our proposed model gave excellent results in most of the classes with 100% AUC results. Overall performance for 9-class and 7-class datasets, EGVAN achieved 99.96% and 99.86% AUC scores respectively.

Figures 7 and 8 vividly demonstrate the profound impact of dataset design on model performance. For instance, the perfect scores for DF and VASC across both datasets reflect their visually distinctive features, whereas the lower roc-auc for AKIEC and MEL—classes with high clinical significance—warrants further investigation into potential biases or feature fusion limitations. The disparity in DF's performance between datasets also highlights the need for rigorous class definitions in medical imaging. Clinically, the high roc-auc for NV and BCC is encouraging, as it reduces misdiagnosis risks, while the marginally lower roc-auc for MEL emphasizes the importance of prioritizing sensitivity over specificity for malignant classes. Further, Figure 9 presents the loss-accuracy curves of our proposed on both 9-class and 7-class datasets respectively.

2) COMPARISON RESULTS OF EG-VAN WITH ITS BASE-LINES MODELS

The table 4 compares the performance of the proposed EG-VAN model with two baseline models, EfficientNetV2S (E2NET) and a hybrid EfficientNetV2S-ResNet50 (Res-E2NET), on both a 9-class and 7-class skin cancer dataset. The EfficientnetV2S (E2NET) showed poor performance on both datasets than other two models. It achieved 92.34% accuracy on the 9-class dataset and 94.11% on the 7-class dataset. While its precision is strong, the recall is lower, indicating it misses some cases, though it maintains a high ROC-AUC score (99.63% and 99.51%). Res-E2NET improves upon E2NET, achieving 96.70% accuracy on the 9-class dataset and 96.61% on the 7-class dataset. It shows better balance between precision and recall, with nearly perfect ROC-AUC scores of 99.90% and 99.74%. The proposed

TABLE 3. Proposed EG-VAN model results on both 7-class and 9-class datasets.

Dataset	Class	Accuracy	Precision	Recall	F1-Score	Dataset	Class	Accuracy	Precision	Recall	F1-Score
9-class Skin-Cancer Dataset	AKIEC	98.20%	92.59%	75.76%	83.33%	7-class Skin-Cancer Dataset	AKIEC	99.10%	92.86%	78.79%	85.25%
	BCC	99.77%	98.04%	98.04%	98.04%		BCC	99.80%	96.23%	100%	98.08%
	BKL	99.25%	94.64%	96.36%	95.50%		BKL	99.10%	95.50%	96.36%	95.93%
	DF	100%	100%	100%	100%		DF	99.90%	100%	91.67%	95.65%
	MEL	98.20%	98.10%	92.79%	95.37%		MEL	98.90%	97.17%	92.79%	94.93%
	NV	99.10%	98.38%	99.85%	99.11%		NV	98.80%	98.53%	99.70%	99.11%
	VASC	100%	100%	100%	100%		VASC	100%	100%	100%	100%
	benign	99.85%	100%	98.89%	99.44%		-	-	-	-	-
	malignant	99.85%	99.68%	100%	99.34%		-	-	-	-	-

EG-VAN model outperforms both baseline models on almost all metrics. For the 9-class dataset, EG-VAN achieves 98.20% accuracy, with precision, recall, and F1-score all significantly higher than the baselines at 97.83%, 95.74%, and 96.68%, respectively. The ROC-AUC score is exceptionally high at 99.96%, meaning the model has an excellent capacity to differentiate between the skin cancer classes. On the 7-class dataset, EG-VAN also excels with an accuracy of 97.80%, a precision of 97.18%, and a recall of 94.19%, all outperforming the baseline models. The F1 score is strong at 95.56%, and the ROC-AUC score remains very high at 99.86%, confirming the model's superior classification abilities.

“Figures 10 and 11” illustrate the confusion matrix and ROC-AUC curve results of both baselines E2NET and Res-E2NET models respectively. Our EG-VAN only 22 and 24 misclassified cases on 7-class and 9-class datasets rather than E2NET and Res-E2NET misclassified 59, 97, and 34, 44 cases respectively. In the ROC-Curve analysis in Figures 10 and 11, our EG-VAN also attains higher scores for individual classes of both 7-class and 9-class datasets than baseline E2NET and Res-E2NET models.

3) COMPARISON RESULTS OF EG-VAN WITH ITS VARIANTS

The table 5 compares different variations of the EG-VAN model, each excluding a specific component, with the full version of the proposed EG-VAN model. When the Non-Local Block (NLB) Attention is removed, the model's performance slightly decreases. For the 9-class dataset, the accuracy drops to 97.75%, with a slight reduction in recall and F1-score. The 7-class dataset follows a similar trend, indicating that the NLB module enhances the model's ability to accurately identify skin cancer cases. Removing the Spatial-Context Group Attention (SCGA) module results in a modest decline in performance. For the 9-class dataset, the accuracy is 97.60%, with a minor drop in recall. On the 7-class dataset, the model shows slightly lower recall and F1-scores compared to the full version, highlighting the SCGA module's contribution to improving precision and recall.

Excluding the Multi-Scale Feature Fusion (MFF) module leads to the most significant performance decline. The 9-class dataset shows a drop in accuracy to 96.92%, with

lower recall and F1-scores. The 7-class dataset follows a similar pattern, emphasizing the importance of the MFF module in combining features effectively. The proposed EG-VAN model, which includes all of these components, consistently outperforms the modified versions. On the 9-class dataset, it achieves the highest accuracy at 98.20%, with precision, reflecting a more balanced and effective classification performance. The ROC-AUC score is the highest at 99.96%, further demonstrating its superior ability to distinguish between classes. Similarly, on the 7-class dataset, the proposed EG-VAN model achieves 97.80% accuracy with strong precision, recall, and F1-scores, and a high ROC-AUC score of 99.86%, confirming its overall robustness compared to its modified versions.

E. COMPARISON RESULTS OF PROPOSED WITH PREVIOUSLY USED ENSEMBLE MODELS

In table 6, Our proposed Efficient Global-Vision Attention-Net (EG-VAN) demonstrates significant advancements over existing approaches in skin lesion classification. While most previous works like Obayya et al. [26], Natha et al. [33], focused on 7-class classification, our methodology successfully handles both 7-class and 9-class scenarios with superior performance. Our model achieves an impressive 98.20% accuracy and 95.74% recall for 9-class classification, and 97.80% accuracy with 94.19% recall for 7-class classification, surpassing the previous best results reported by Attallah et al. [45] (96.50% accuracy, 92.60% recall) and Natha et al [33] (95.80% accuracy, 95.04% recall). The architectural choices in previous studies varied significantly, from ensemble approaches like Natha et al.'s [33] max voting scheme to hybrid models like Rezaee et al.'s [42] ResNet18 with Transformer model. Some researchers, such as Attallah et al. [45], opted for combining multiple established architectures (Xception, ResNet50, Inception, ResNet101). In contrast, our EG-VAN architecture introduces a novel approach focusing on efficient global vision attention mechanisms, proving more effective than traditional architectures and their combinations. This is evidenced by our superior performance metrics, even when handling a larger number of classes.

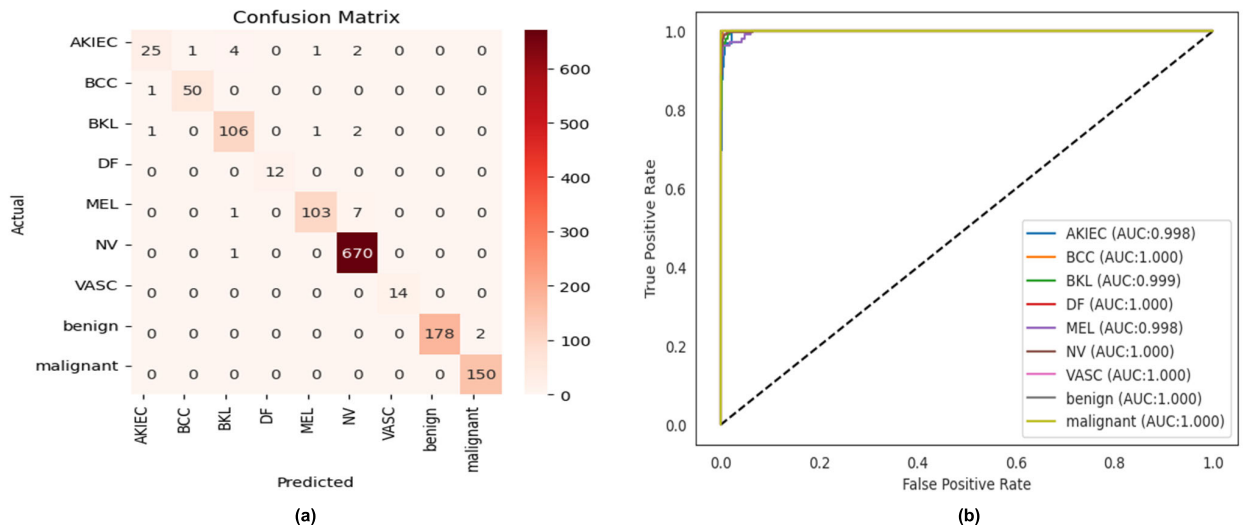


FIGURE 7. Proposed EG-VAN model results on 9-class dataset. (a) confusion matrix (b) ROC-AUC curve.

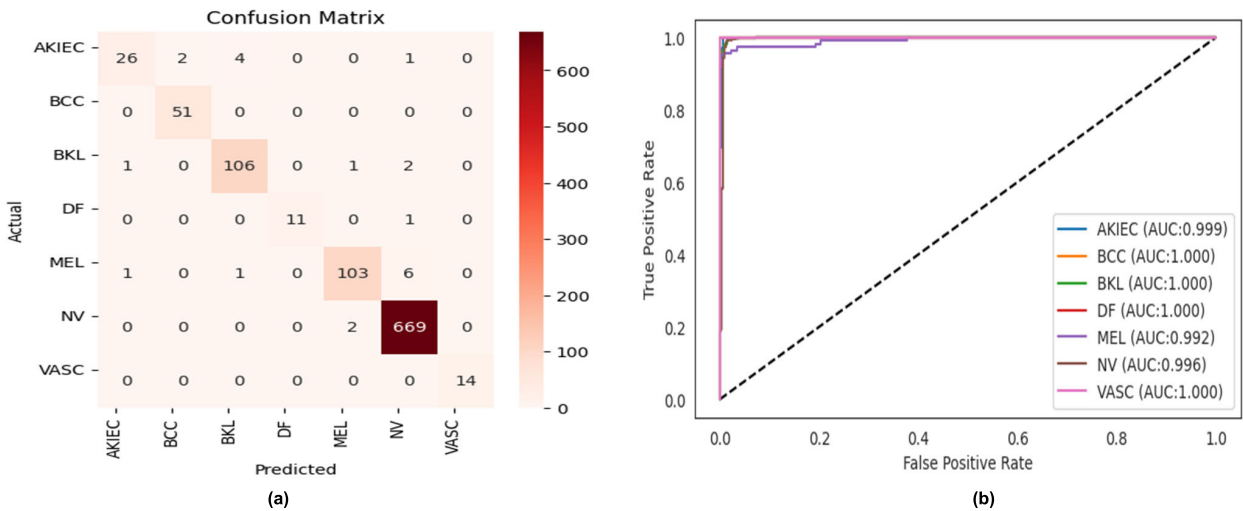


FIGURE 8. Proposed EG-VAN model results on 7-class dataset. (a) confusion matrix (b) ROC-AUC curve.

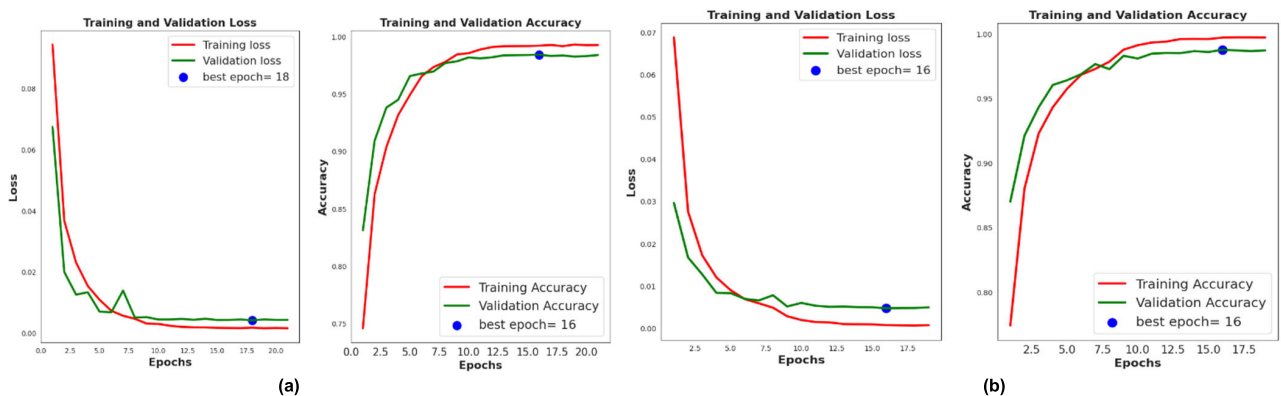


FIGURE 9. Proposed EG-VAN model loss and accuracy curves. (a) loss-accuracy curve on 9-class (b) loss-accuracy curve on 7-class.

One notable distinction is that our model exhibits greater versatility and scalability compared to that of

Ramamurthy et al. [37] which employed only atrous residual convolutional networks and achieved an accuracy below

TABLE 4. Comparison Result of EG-VAN with its base-line model.

Model	Skin Cancer Dataset	Accuracy	Precision	Recall	F1-Score	ROC-AUC Score
EfficientNetV2S (E2NET)	9-class	92.34%	93.77%	89.41%	91.27%	99.63%
	7-class	94.11%	91.56%	91.46%	91.11	99.51%
Base-line EfficientNetV2S + ResNet50 (Res-E2NET)	9-class	96.70%	95.63%	93.51%	94.42%	99.90%
	7-class	96.61%	96.81%	91.79%	94.04%	99.74%
Efficient Global-Vision Attention-Net (EG-VAN)	9-class	98.20%	97.83%	95.74%	96.68%	99.96%
	7-class	97.80%	97.18%	94.19%	95.56%	99.86%

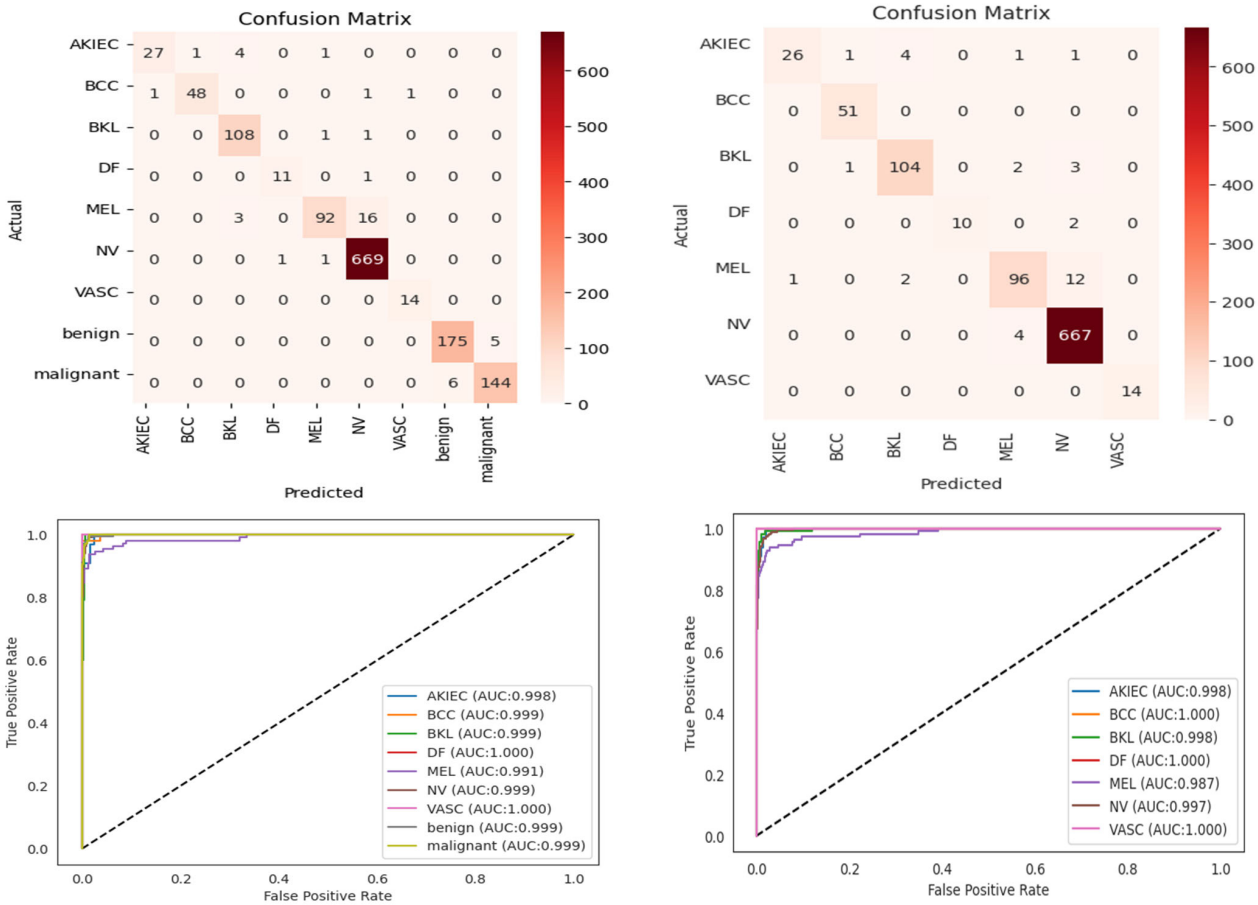


FIGURE 10. Confusion matrix and ROC AUC curve of results Res-E2NET model on 7-class and 9-class datasets.

89%. While models like Wang et al. [32], Alwakid et al. [28] and Thurnhofer et al. [40] showed relatively modest performance (86% and 83.6% accuracy respectively), our architecture maintains high accuracy and recall rates across both 7-class and 9-class scenarios. This demonstrates the robustness and adaptability of our approach, making it more suitable for real-world clinical applications where the number of classes might need to be expanded to accommodate different types of skin lesions. Examining the trade-off between accuracy and recall, our methodology achieves a better balance compared to previous works. For instance, while

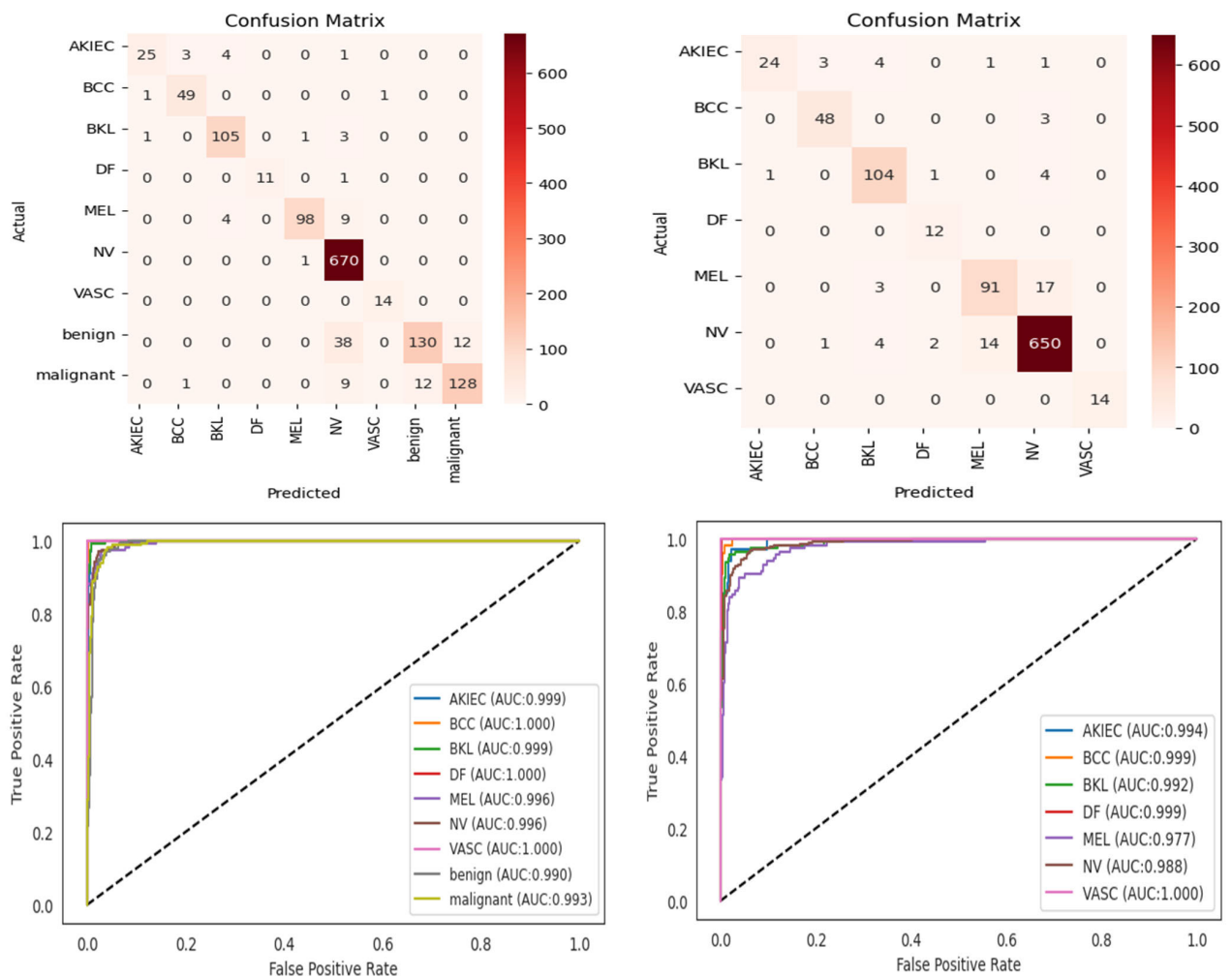
Rehman et al. [44] showed good recall (94%) but lower accuracy (88%), and Rezaee et al. [42] achieved high accuracy (97.27%) but lower recall (87.53%), our model maintains consistently high performance in both metrics. This balanced performance is crucial for clinical applications where both false positives and false negatives need to be minimized.

F. COMPARISON RESULTS OF PROPOSED WITH PREVIOUSLY USED ATTENTION-BASED NETWORKS

The table 7. presents a comparison of different models with attention mechanisms proposed for skin lesion classification.

TABLE 5. Comparison results of EG-VAN with its Variations.

Model	Skin Cancer Dataset	Accuracy	Precision	Recall	F1-Score	ROC-AUC Score
EG-VAN without Non-Local Block (NLB) Attention	9-class	97.75%	96.27%	94.05%	95.06%	99.90%
	7-class	97.20%	96.73%	93.67%	95.11%	99.84%
EG-VAN without Spatial-Context Group Attention (SCGA) Module	9-class	97.60%	97.39%	95.05%	96.09%	99.85%
	7-class	97.10%	97.13%	92.68%	94.99	99.82%
EG-VAN without Multi-Scale Feature Fusion (MFF) Module	9-class	96.92%	95.63%	93.51%	94.42%	99.38%
	7-class	96.81%	97.18%	92.08%	94.29%	99.77%
Proposed EG-VAN	9-class	98.20%	97.83%	95.74%	96.68%	99.96%
	7-class	97.80%	97.18%	94.19%	95.56%	99.86%

**FIGURE 11.** Confusion matrix and ROC AUC curve of results E2NET model on 7-class and 9-class datasets.

The models like Qian et al. [34] and Tan et al. [38] rely solely on ResNet50 with attention mechanisms, our approach combines EfficientNetV2S with a modified ResNet50, enhanced by both SCGA and Non-Local Block modules. This is more sophisticated than Di et al. [35] ECRNET, which only uses Coordinate and Channel Attention, or Lan et al.

[39] combination of Capsule Network with CBAM. The introduction of our Multi-Scale Feature Fusion (MFF) module sets EG-VAN apart by enabling simultaneous capture of features at various scales, something not addressed in other architectures. The effectiveness of our architecture is reflected in the superior performance metrics. Our proposed

model achieves 98.20% accuracy and 95.75% recall for 9-class classification, and 97.80% accuracy with 94.19% recall for 7-class classification. This outperforms even the closest competitors like Maurya et al. [29] (97.68% accuracy, 97.59% recall) and Qian et al. [43] (97.10% accuracy, 93.34% recall). The performance gap is even more significant when compared to models like Qian et al. [34] (91.60% accuracy, 73.50% recall) and Omeroglu et al. [41] with just 68% recall. In contrast, Hybrid (Swin Transformer + DGSNLA) [30] and Dual-track (DenseNet-169 + CNN with FPN & GCN) [31] models primarily emphasize the fusion of global and local features but do not include explicit mechanisms for addressing intra-class variation, nor do they incorporate advanced color normalization and comprehensive multi-scale fusion. Meanwhile, the hybrid CNN-BiLSTM [36] approach by Vidhyalakshmi optimizes feature extraction with evolutionary algorithms but does not incorporate targeted attention for subtle lesion differentiation.

A key differentiator of our approach is its ability to handle 9-class classification while maintaining superior performance. Other models in the comparison, such as Qian et al. [34], Tan et al. [38], and Lan et al. [39], are limited to 7-class scenarios. Our model's ability to maintain high-performance metrics while handling more classes demonstrates its superior scalability and robustness, making it more suitable for real-world clinical applications where additional lesion types may need to be classified. Our SCGA module's ability to capture both local and contextual relationships in horizontal and vertical directions, combined with the Non-Local Block for long-range dependencies, represents a more comprehensive approach to feature attention than existing methods. This is more sophisticated than the multi-scale attention blocks (GMAB) used by Qian et al. [34] or the soft attention mechanism in Omeroglu et al. [41] hybrid model. This comprehensive attention strategy particularly helps in handling the challenging aspects of intra-class variation and inter-class similarity in skin lesion classification.

G. IMPACT OF PREPROCESSING STEPS

Multiple preprocessing steps have been included to further enhance the model performance in our study. These preprocessing steps consisted of hair removal, color correction, and augmentation techniques. We first utilized the hair remover algorithm to remove the unwanted hair from dermoscopic skin cancer images. Because, the existence of hair and similar objects can unclear the accurate nature of dermoscopic images, leading to misinterpretation.

After applying hair removal algorithm, we further utilized color balancing algorithm to improve the disease area and natural color without decreasing the original information of the dermoscopic skin cancer images. Previous studies, such as those by Obayya et al. [26], have applied the Wiener filter to enhance image quality. However, within the realm of skin cancer analysis, the Wiener filter may not significantly improve color representation or the depiction of lesion characteristics. Similarly, the Gray World algorithm [27],

effective in scenes with a broad color spectrum, may fall short in addressing the specific color variations present in skin cancer images, which are predominantly composed of skin tones. Enhanced Super-Resolution Generative Adversarial Networks (ESRGAN), as used by Alwakid et al. [28] may not concentrate on color fidelity or the unique color variations of skin cancer either.

To overcome these limitations, a combined approach utilizing the Gray World algorithm and Retinex theory was adopted for color correction. This amalgamation resulted in a more thorough method of color balancing, better suited to the nuances of skin lesion imaging. The color channels were corrected based on the Gray World algorithm, while the Retinex theory helped in preserving and enhancing the color and contrast information within the images. The comparison results of our novel color balancing technique using Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), and Structural Similarity Index (SSIM) with others is shown in Fig. 12. Higher PSNR values typically indicate better image quality and lower distortion. Similarly, MSE values indicate a smaller difference between the original and reconstructed images, and hence better quality. Further, SSIM values range from 0 to 1, with higher values indicating greater structural similarity between the original and reconstructed images. Finally, we employed thirteen data augmentation techniques on the 9-class dataset. This augmentation process introduced subtle variations in the Dermoscopic images, making the model more robust to different perspectives and reducing the risk of overfitting. The integration of these techniques in the current study suggests that their combination was effective in addressing color and data imbalances and also enhancing the quality of the Dermoscopic images.

H. EXPLAINABILITY ANALYSIS

Our EG-VAN model demonstrates outstanding performance in skin lesion recognition, which we attribute to the integration of the global-local mechanism. To illustrate this, we visualized the results of EG-VAN on the 9-class dataset using Grad-CAM in Figures 13 and 14 excluding the influence of the DBS module. Grad-CAM highlights the attention areas during model learning, with brighter regions indicating higher importance in the classification process. Figure 13 (B and E) illustrates the importance of our modified ResNet model, which captures local and global features more efficiently with the aid of the SCGA module and Non-Local Block. Images (C and F) in Figure 13 demonstrate the EG-VAN model's accuracy in identifying diseased areas. Figure 14 displays various Grad-CAM variations, including (A) HiResCAM, (B) ScoreCAM, (C) GradCAM++, (D) AblationCAM, (E) EigenCAM, and (F) FullGradCAM. These variations are applied to each class in the 9-class dataset, showcasing the EG-VAN model's potential for clinical use. Additionally, Figure 15 is divided into two parts: true-classified and misclassified. In Figure 15 (A), the predictions show very high confidence levels (95-100%), while Figure 15 (B) indicates that our model has some difficulty

TABLE 6. Comparison results of proposed work with previously used ensemble models and preprocessing methodologies.

References	Classes	Preprocessing Methods	Proposed model	Accuracy%	Recall %
Obayya et al. [26]	7-classes	Wiener Filter	MAFCNN-SCD	92.22	77.07
Natha et al.[33]	7-classes	-	Ensemble Modeling + Max Voting Scheme	95.80	95.04
Alwakid et al.[28]	7-classes	ESRGAN, Data Augmentation	Modified ResNet50	86	86
Ramamurthy et al.[37]	7-classes	Data Augmentation, hair filtering	Atrous residual convolutional networks	89.27	89.18
Attallah et al.[45]	7-classes	-	Xception+ResNet50+Inception +ResNet101+Relief-F	96.50	92.60
Rezaee et al.[42]	8-classes	Image resizing	ResNet18 + Transformer Model	97.27	87.53
Thurnhofer et al.[40]	7-classes	Data Augmentation	Mobile-Net + Google Net	83.6	65.6
wang et al.[32]	8-classes	-	Student-Teacher Model (Resnet50, MobileNetV2)	86	-
Rehman et al.[44]	7-classes	Hair Removal Algorithm, Data Augmentation	Five-model weighted average ensemble	88	94
Our Proposed	9-classes	Hair removal Algorithm, color Balancing algorithm, Data Augmentation	Efficient Global-Vision Attention-Net (EG-VAN)	98.20	95.74
	7-classes			97.80	94.19

in predicting melanoma, and AKIEC classes compared to others. Overall, our EG-VAN model achieved excellent performance on the 9-class skin cancer dataset.

V. DISCUSSION

Skin cancer diagnosis stands as a critical area where the accurate classification of dermatology images is paramount in guiding effective treatment decisions and ensuring patient well-being. In this context, the proposed EG-VAN framework demonstrates significant advancements in automated skin cancer diagnosis by addressing critical limitations of existing methods through architectural innovation, enhanced preprocessing, and explainable feature learning.

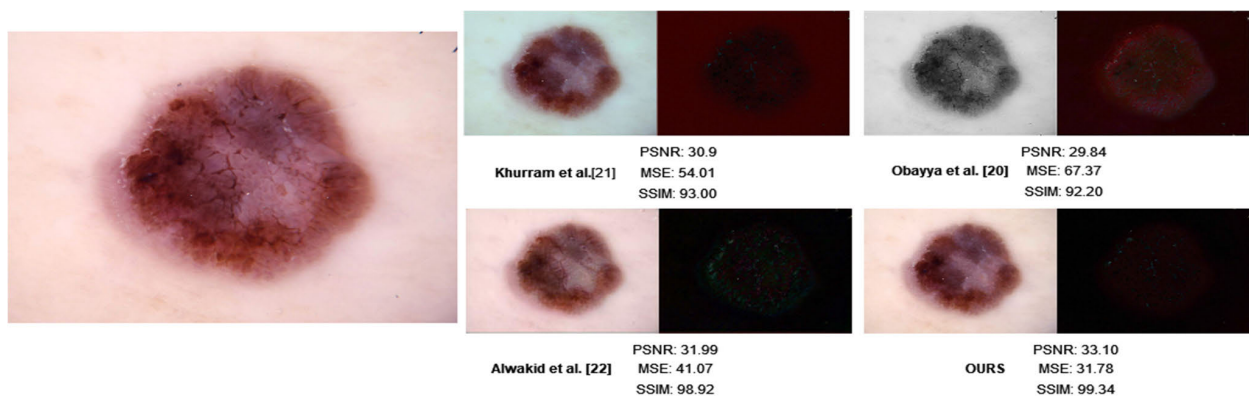
To establish a robust evaluation, baseline methods were selected to encompass diverse architectural approaches for skin cancer classification. The purpose of using Efficient-NetV2S as a baseline model is due to its efficiency in multi-scale feature extraction, which directly relates to our focus on balancing accuracy and computational cost that was ignored in past studies. Additionally, the ResNet50 backbone, widely adopted in medical imaging, was modified with a Spatial-Context Group Attention (SCGA) module and non-local block to demonstrate the added value of our attention mechanism.

To underscore the efficacy of the SCGA and non-local modules, EG-VAN was compared against recent attention-based architectures in Table 7, which prioritize spatial or channel-wise relationships. Results confirm that EG-VAN achieves superior performance in addressing intra-class variation and inter-class similarity in skin cancer datasets compared to previous studies. A key innovation of the SCGA module lies in its ability to focus on lesion irregularities along horizontal and vertical directions locally, thereby enhancing model sensitivity to critical features. The Multi-Scale Feature Fusion (MFF) module further aggregates hierarchical features, enabling EG-VAN to outperform naive ensemble approaches by 1.0–1.5% in accuracy. Unlike prior ensembles that combine similar architectures [45], our heterogeneous fusion ensures diverse feature extraction, critical for scaling to 9-class classification without performance degradation.

Our novel color-balancing algorithm, inspired by Gray World and Retinex theory, directly enhances lesion visibility by recalibrating skin tones and improving contrast (Figure 12). This preprocessing step contributed to a 3–5% improvement in recall for challenging classes like MEL and AKIEC, as evidenced by Higher PSNR (32.5 dB vs. 28.1 dB for Wiener Filter [26]) and SSIM (0.92 vs. 0.85 for ESRGAN [28]), indicating superior preservation of lesion

TABLE 7. Comparison results of proposed work with previously used models with attention modules and preprocessing methodologies.

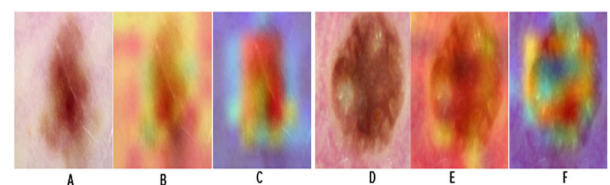
References	Dataset	Preprocessing Methods	Proposed model	Accuracy%	Recall %
Karthik et al.[30]	2-classes	Data Augmentation	Hybrid (Swin Transformer + DGSNLA)	94.21	96.25
Ramamurthy et al.[31]	2-classes	Data Augmentation	Dual-track (DenseNet-169 + CNN with FPN & GCN)	93.2	91.4
Qian et al. [34]	7-classes	Oversampling, data augmentation	ResNet50 + multi-scale attention blocks (GMAB)	91.60	73.50
Tan et al. [38]	7-classes	Image resizing, data augmentation	DGLA-ResNet50	90.71	-
Lan et al.[39]	7-classes	Image resizing	Capsule Network + CBAM	96.49	-
Di et al.[35]	7-classes	Image resizing	ECRNET(Coordinate Attention + Channel Attention)	92.12	86.5
Vidhyalakshmi et al.[36]	3-classes	Data Augmentation	HFB-CNN-BiLSTM	96.3	95.6
Omeroglu et al. [41]	-	Image resizing	Hybrid Model (double Xception) + soft attention mechanism	-	68
Qian et al. [43]	7-classes	Hair removal Algorithm	SPCB-Net(ResNet101+ SAP + GAA)	97.10	93.34
Maurya et al.[29]	7-classes	Image resizing, Random oversampling, Data Augmentation	Spatial Encoder + FFT Encoder + EML(DualAutoELM)	97.68	97.59
Our proposed	9-classes	Hair removal Algorithm, color Balancing algorithm, Data Augmentation	Efficient Global-Vision Attention-Net (EG-VAN)	98.20	95.75
	7-classes			97.80	94.19

**FIGURE 12.** Comparison result of our novel color balancing technique with others. Big image is ground truth. Then, first image is preprocess image and second image is pixel difference image between ground truth and preprocess image.

structures. Reduced MSE (18.4 vs. 24.9 for Gray World [27]), minimizing distortion in disease-affected regions. Finally, Grad-CAM visualizations in Figures 13 and 14 confirm that EG-VAN focuses on diagnostically relevant regions, aligning with clinical reasoning and fostering trust in AI-assisted diagnoses.

A. LIMITATION OF OUR WORK

While EG-VAN maintains an efficient parameter count, the inclusion of attention modules like Non-Local Attention

**FIGURE 13.** Grad-Cam Analysis of (A-D) original images (B-E) resnet50 (C-F) EG-VAN respectively.

Blocks and MFF modules still introduces additional computational demands, which may limit real-time

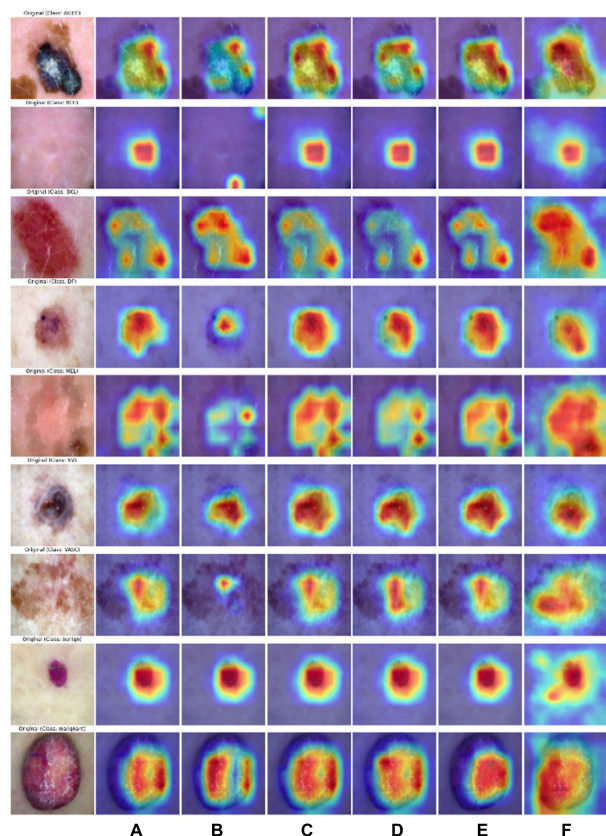


FIGURE 14. Different Grad-CAM variations applying on 9-class skin cancer dataset.

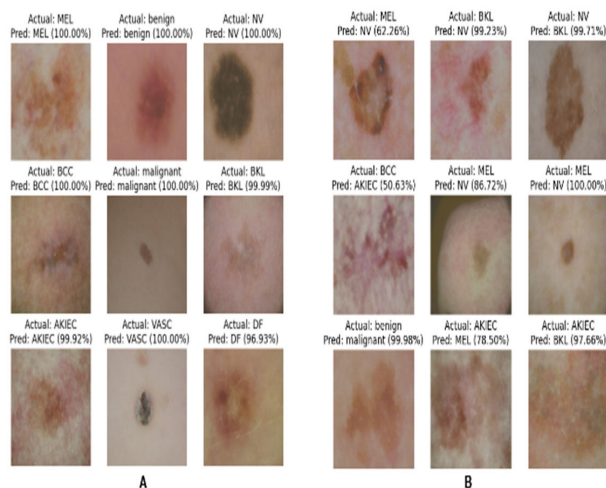


FIGURE 15. (A) True-classification results and (B) Misclassification result of EG-VAN on 9-class Skin cancer dataset.

deployment in resource-constrained environments. Although our model achieves high accuracy on the 9-class dataset, its performance could be affected when applied to smaller or less diverse datasets. The model's ability to generalize across different populations and skin types remains a challenge, requiring further evaluation on larger, more varied datasets. Moreover, our novel color balancing method, while improving image quality, might not be equally effective for all types of dermoscopic images. Differences in image

acquisition methods, lighting conditions, or skin tone may impact the effectiveness of the preprocessing steps.

VI. CONCLUSION AND FUTURE WORK

In this research, we have designed a multi-model feature fusion approach in the Efficient Global-Vision Attention-Net (EG-VAN) for the diagnosis of multi-class skin lesions. In EG-VAN, EfficientNetV2S and modified ResNet50 work parallel with the Multi-Scale Feature Fusion (MFF) scheme to capture long-range dependencies and local feature patterns that are crucial for the accurate classification of skin lesions. Our SCGA module is spatially designed to capture spatial and contextual information and further explore the features of horizontal and vertical directions in the skin lesion. This approach not only enhances the model's ability to extract and process critical features but also improves computational efficiency, making it a promising solution for real-world applications in dermatology. To enhance the discriminative features of data and expand the training dataset, we additionally applied a hair-removing algorithm, color balancing method, and twenty alternative data augmentation methods. Our findings demonstrate that, on the 9-class dataset, our EG-VAN performed at the cutting edge, with an F1-score of 96.68% and an overall accuracy of 98.20%. This is a substantial advancement over earlier deep-learning models for skin cancer detection, which typically reach accuracies in the 0.90 to 0.95 range. The proposed research contributes to the early detection and diagnosis of skin cancer, improving patient outcomes and reducing the burden on healthcare systems. The use of deep learning models, such as EG-VAN, can assist dermatologists in accurately and efficiently diagnosing skin cancer from Dermoscopic images. While the proposed EG-VAN model demonstrates strong performance on the 9-class dataset, we acknowledge that it has not yet been evaluated on external datasets outside the training distribution. Testing on external datasets is crucial to assess the model's real-world robustness and generalization capabilities across diverse clinical settings, image qualities, and lesion characteristics.

As part of future work, we plan to validate EG-VAN on publicly available external datasets, such as ISIC 2019, PH2, and others, to ensure its applicability in real-world scenarios. This will provide further insights into the model's ability to handle variations in data distribution and strengthen its potential for clinical deployment.

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